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M168_Notes_2.4

2.4 Matrices

A matrix is a compact way of organizing and displaying data.

A matrix is often denoted by a capital letter M or A.

A matrix having m rows and n columns is an $m \times n$ matrix

Shape - $m \times n$

$$M_1 = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$$ is a 3x1 matrix

$$M_2 = \begin{bmatrix} 1 & -1 & 2 \end{bmatrix}$$ is a 1x3 matrix

These two matrices are NOT EQUAL.

A matrix is called square if it has the same number of rows and columns.

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$ is a 2 x 2 square matrix.

$a_{12} = 2$

$m \times n$
if $m = n$

$\boxed{a_{ij}}$ is the element in the ith row and j^{th} column of matrix A.

$A = [a_{ij}]_{m \times n}$

a_{ij} will stand for (i,j)-th entry of the matrix

Example –

There are three stores. In the first week store I sold 88 loaves of bread, 48 quarts of milk, 16 jars of peanut butter and 112 pounds of cold cuts. At the same time, store II sold 105 loaves of bread, 72 quarts of milk, 21 jars of peanut butter and 147 pounds of cold cuts. Store III sold 60 loaves of bread, 40 quarts of milk, 0 jars of peanut butter and 50 pounds of cold cuts.

Organize this data in a 3 x 4 matrix.

$$\begin{array}{c} \\ \text{Store-I} \\ \text{Store-II} \\ \text{Store-III} \end{array} \begin{array}{c} \begin{array}{cccc} B & M & PB & CC \end{array} \\ \begin{bmatrix} 88 & 48 & 16 & 112 \\ 105 & 72 & 21 & 147 \\ 60 & 40 & 0 & 50 \end{bmatrix} \end{array}$$

MATRIX ALGEBRA

Equality - two matrices are equal if and only if each pair of corresponding elements are equal.

Example - Find the values of a, b, c, d given

$$\begin{bmatrix} 1 & b \\ 3 & 0 \end{bmatrix} = \begin{bmatrix} a & -1 \\ c & d \end{bmatrix}$$

Same shape
respective entries must be equal.
$1=a,\ b=-1,\ 3=c,\ 0=d$

Addition - two matrices are added by adding the pairs of elements in each location.

Example - find A+B where $A = \begin{bmatrix} 2 & -3 \\ 0 & 5 \\ .25 & 6 \end{bmatrix}$ and $B = \begin{bmatrix} 0.25 & 7 \\ -3 & 1.5 \\ 9 & 2 \end{bmatrix}$

To add Same shape.

$$A+B = \begin{bmatrix} 2.25 & 4 \\ -3 & 6.5 \\ 9.25 & 8 \end{bmatrix}$$

Transpose - The transpose of a matrix is found by switching the rows and columns of the matrix.

If A is a 3x2 matrix then A^T will be a 2x3 matrix.

$$A = \begin{bmatrix} 2 & -3 \\ 0 & 5 \\ 0.25 & 6 \end{bmatrix}_{3\times 2} \qquad A^T = \begin{bmatrix} 2 & 0 & 0.25 \\ -3 & 5 & 6 \end{bmatrix}_{2\times 3}$$

Scalar multiplication –

A scalar is a number (NOT a matrix).

Multiply a matrix by a scalar by multiplying every element in the matrix by the scalar.

Example - find $-2A$.

$$2A = 2\begin{bmatrix} 2 & -3 \\ 0 & 5 \\ 0.25 & 6 \end{bmatrix}_{3\times 2} = \begin{bmatrix} 4 & -6 \\ 0 & 10 \\ 0.5 & 12 \end{bmatrix}_{3\times 2}$$